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EXPERIMENT - 5

IPTables Firewall Configuration in Kali Linux

Aim:
Toperform and analyze various firewall configurations using IPTables in Kali Linux to control network traffic and enhance system security.

Procedure:

Step 1: Start Environment
Open Oracle VirtualBox and start Kali Linux virtual machine. Log in to the system.

Step 2: Open Terminal and Check IPTables
Open the terminal and verify that IPTables is installed and available in the system.

Step 3: View Existing Rules
Check the current firewall rules to understand the default configuration before making changes.

Step 4: Configure Firewall Rules
Set default policies to control incoming and outgoing traffic.
Allow necessary services such as localhost communication, SSH, HTTP, and HTTPS.
Block unwanted or suspicious IP addresses.

Step 5: Save and Verify Rules
Save the configured firewall rules and verify that they are applied correctly by reviewing the rule list.

Output:

```
student@kali: ~
Session  Actions  Edit  View  Help

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport22 -j ACCEPT
iptables v1.8.11 (nf_tables): unknown option "--dport22"
Try `iptables -h' or 'iptables --help' for more information.

┌──(student㉿kali)-[~]
└─$ sudo iptables -S
-P INPUT DROP
-P FORWARD ACCEPT
-P OUTPUT ACCEPT
-A INPUT -i lo -j ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 22 -j ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 80 -j ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -p tcp --dport 443 -j ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -s 192.168.1.100 -j DROP

┌──(student㉿kali)-[~]
└─$ sudo iptables -L
Chain INPUT (policy DROP)
```

```
student@kali: ~
Session  Actions  Edit  View  Help
└─$ john --show hash.txt
0 password hashes cracked, 0 left

┌──(student㉿kali)-[~]
└─$ iptables --version
iptables v1.8.11 (nf_tables)

┌──(student㉿kali)-[~]
└─$ sudo iptables -L
[sudo] password for student:
Chain INPUT (policy ACCEPT)
target     prot opt source               destination

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination

┌──(student㉿kali)-[~]
└─$ sudo iptables -P INPUT DROP

┌──(student㉿kali)-[~]
└─$ sudo iptables -P OUTPUT ACCEPT

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -i lo -j ACCEPT
```

```
student@kali: ~
Session  Actions  Edit  View  Help
    0     0 DROP       all  --  *      *       192.168.1.100        0.0.0.0/0

┌──(student㉿kali)-[~]
└─$ sudo iptables -L INPUT -n | grep DROP
Chain INPUT (policy DROP)
DROP       all  --  192.168.1.100        0.0.0.0/0

┌──(student㉿kali)-[~]
└─$ sudo iptables-save > firewall_rules.txt

┌──(student㉿kali)-[~]
└─$ cat firewall_rules.txt
# Generated by iptables-save v1.8.11 (nf_tables) on Tue Feb  3 16:21:07 2026
*filter
:INPUT DROP [4:288]
:FORWARD ACCEPT [0:0]
:OUTPUT ACCEPT [4:288]
-A INPUT -i lo -j ACCEPT
-A INPUT -p tcp -m tcp --dport 22 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 80 -j ACCEPT
-A INPUT -p tcp -m tcp --dport 443 -j ACCEPT
-A INPUT -s 192.168.1.100/32 -j DROP
COMMIT
# Completed on Tue Feb  3 16:21:07 2026

┌──(student㉿kali)-[~]
└─$
```

```
student@kali: ~
Session  Actions  Edit  View  Help

┌──(student㉿kali)-[~]
└─$ sudo iptables -A INPUT -s 192.168.1.100 -j DROP

┌──(student㉿kali)-[~]
└─$ sudo iptables -L
Chain INPUT (policy DROP)
target     prot opt source               destination
ACCEPT     all  --  anywhere             anywhere
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:ssh
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:http
ACCEPT     tcp  --  anywhere             anywhere             tcp dpt:https
DROP       all  --  192.168.1.100        anywhere

Chain FORWARD (policy ACCEPT)
target     prot opt source               destination

Chain OUTPUT (policy ACCEPT)
target     prot opt source               destination

┌──(student㉿kali)-[~]
└─$ sudo iptables -L -v -n
Chain INPUT (policy DROP 4 packets, 288 bytes)
 pkts bytes target     prot opt in     out     source               destinati
on
    0     0 ACCEPT     all  --  lo     *       0.0.0.0/0            0.0.0.0/0

    0     0 ACCEPT     tcp  --  *      *       0.0.0.0/0            0.0.0.0/0
```

Result

Various firewall rules were implemented using IPTables, demonstrating how network traffic can be controlled to enhance system security.